



United International University
Department of Computer Science and Engineering
Final Term Examination || Trimester: Summer – 2025
School of Science and Engineering
Course: Math -2205 (Probability & Statistics)
Total marks – 40 || Duration – 2 hours

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

- Q1** (a) During a software testing phase, each module has a 0.1 probability of containing a critical bug. If a tester randomly and independently selects 12 modules, find the probability that exactly 3 modules have critical bugs. [2]
- (b) A network server receives an average of 4 data requests per second. Assuming requests arrive randomly and independently, find the probability that in two seconds the server receives at least 1 request. [3]
- (c) The processing time for a batch of transactions on a data server is normally distributed with a mean of 2.5 seconds and a standard deviation of 0.4 seconds. Find the probability that a transaction takes less than 2 seconds to process. [3]
- (d) If $X \sim N(\mu, \sigma^2)$, then calculate $P(\mu - \sigma < x < \mu + \sigma)$ [2]
- Total [10 marks]**

- Q2** The time (in minutes) taken by a computer to complete a data backup operation is modeled as a continuous random variable X with the probability density function:

$$f(x) = \begin{cases} k(10 - x), & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Determine the value of k that makes $f(x)$ a valid probability density function. [2]
- (b) Find the Cumulative Distribution Function (CDF) $F(x)$. [2]
- (c) Find the probability that the backup takes less than 5 minutes. [2]
- (d) Compute the expected time for the backup to complete. [2]
- (e) Determine the median backup time. [2]
- Total [10 marks]**

- Q3** (a) A random sample of 200 students is selected, and 126 of them report that they prefer online classes. Construct a 95% confidence interval for the true population proportion. [2]
- (b) (i) A company claims that the average typing speed of its data entry clerks is at least 60 words per minute (wpm) after training. A supervisor suspects that the training program has actually improved performance and wants to test whether the mean speed is greater than 60 wpm. A random sample of 40 clerks shows a mean typing speed of 62.3 wpm with a standard deviation of 4.8 wpm. At the 5% significance level, test whether the average speed has increased. [5]
- (ii) Define Type I and Type II error [2]
- (c) A company produces energy-saving LED bulbs that are supposed to last at least 10,000 hours on average. To ensure quality, the quality control department tests a random sample of bulbs. Define what a Type I error would mean in this context. [1]
- Total [10 marks]**

- Q4** (a) A factory has **two machines** producing the same metal bolt:
- Machine A produces 60% of the bolts. Its defect rate is 3%. Machine B produces 40% of the bolts. Its defect rate is 5%. You pick one bolt at random from the daily production.

P.T.O

(i) What is the probability that the chosen bolt is defective? [1]

(ii) If the chosen bolt is found to be defective, what is the probability that it came from Machine B? [2]

(b) A delivery company gives drivers a daily performance bonus based on the number of successful deliveries made in a day. The number of deliveries completed varies due to traffic, weather, and demand. The company's data show the following probability distribution for the number of deliveries a driver completes in a day, and the corresponding bonus earned.

Number of deliveries	5	6	7	8	9
Probability	0.1	0.2	0.4	0.2	0.1
Bonus earned (in Tk)	200	250	300	350	400

(i) Find the expected number of deliveries a driver completes per day. [1]

(ii) Find the standard deviation of the daily bonus (in Tk.) a driver earns. [3]

(c) A survey was conducted among 200 university students to study the relationship between owning a laptop and subscribing to online learning platforms. The findings are summarized below:

Category	Own a laptop	Do not Own a laptop	Total
Subscribe online	80	60	140
Do not subscribe	20	40	60
Total	100	100	200

A is the event that a student owns a laptop; **B** is the event that a student subscribes to online learning. [2]

(i) Find the P(A) and P(B) [1]

(ii) Test whether events A and B are independent

Total [10 marks]

Formulae:	
Confidence Interval population mean	: $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$
Confidence Interval population Proportion	: $\hat{p} - z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < p < \hat{p} + z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Probability distribution function for Binomial Distribution	: $f(x) = nC_x p^x (1-p)^{n-x}; x = 0, 1, 2, \dots, n$
Probability distribution function for Poisson Distribution	: $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$
Probability distribution function for Normal Distribution	: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$